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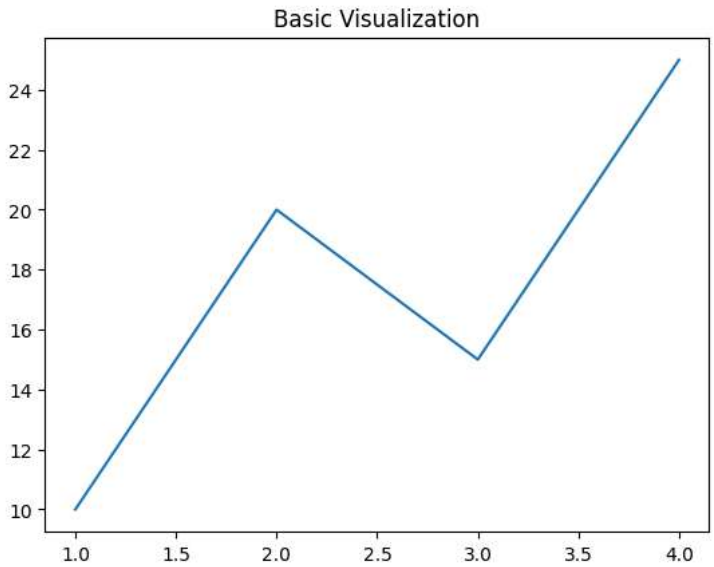
Aim: Data Visualization with Matplotlib: Data visualization on a dataset using matplotlib and seaborn libraries • Scatter plot • Line plot • Bar plot • Histogram • Box plot • Pair plot

1 . What is the importance of data visualization in data analysis? How does visualization help in exploratory data analysis (EDA)?

ans:-

Data visualization is important in data analysis because it transforms complex and large datasets into visual formats that are easy to understand and interpret. It helps analysts quickly identify patterns, trends, relationships, and anomalies that may not be visible in raw data. In exploratory data analysis (EDA), visualization plays a key role by enabling the detection of missing values, outliers, and data distribution. Graphs like scatter plots, histograms, and box plots make it easier to explore the structure of the data and support better decision-making and hypothesis formation.

```
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [10,20,15,25]
plt.plot(x,y)
plt.title("Basic Visualization")
plt.show()
```

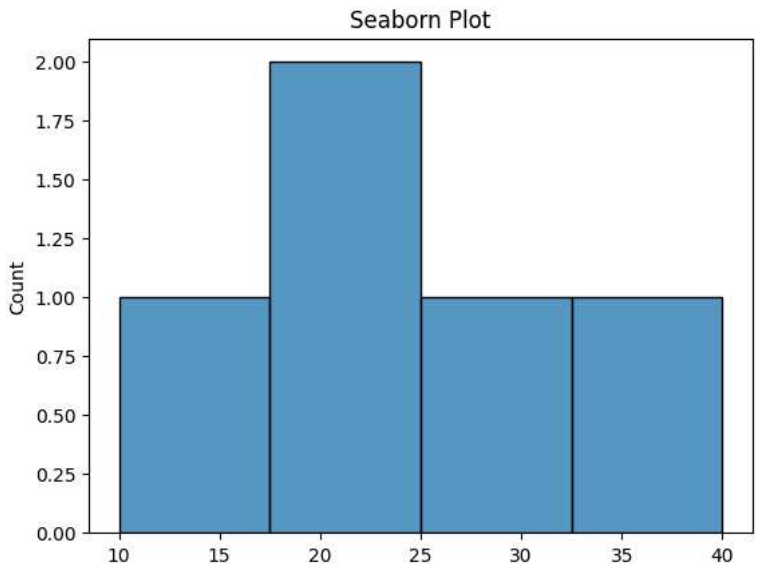



Q2 . What is the difference between Matplotlib and other visualization libraries?

ANS:-

Matplotlib is a fundamental data visualization library in Python used to create static, animated, and interactive plots. It provides detailed control over every element of a graph, but requires more coding. Other libraries like Seaborn and Plotly are built on top of Matplotlib and simplify the process of visualization by providing befler default styles and advanced features. Thus, Matplotlib is mainly used when detailed customization is needed, while other libraries are preferred for quick and visually appealing plots. Matplotlib → Flexible and highly customizable Seaborn → Simple and ideal for statistical visualization Plotly → Interactive and web-based graphs

```
import seaborn as sns
import matplotlib.pyplot as plt
data = [10,20,20,30,40]
sns.histplot(data)
plt.title("Seaborn Plot")
plt.show()
```

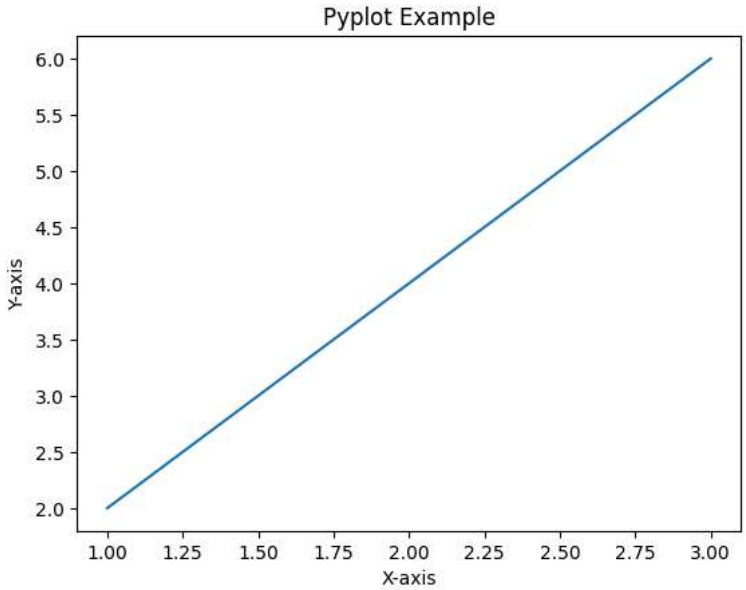



Q3. What is the purpose of the pyplot module in Matplotlib?

ANS:-

The pyplot module is a collection of functions in Matplotlib that provides a simple interface for creating and managing plots. It allows users to create figures, plot data, and customize graphs step by step. Pyplot is widely used because it follows a procedural approach similar to MATLAB, making it easy for beginners to understand. It also helps in controlling various components of a graph such as axes, titles, labels, legends, and grid. Thus, pyplot acts as the main tool for visualization in Matplotlib.

```
import matplotlib.pyplot as plt
plt.plot([1,2,3], [2,4,6])
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Pyplot Example")
plt.show()
```

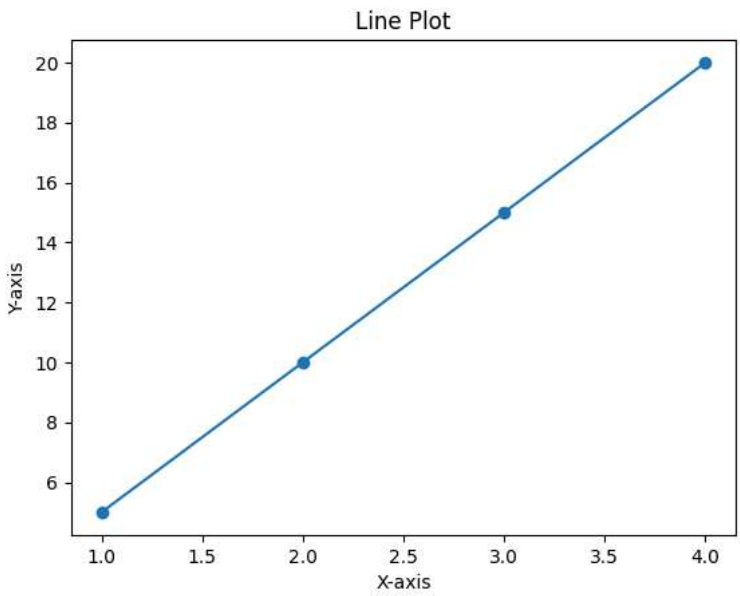



Q4. Explain the process of creating a basic line plot.

ANS:-

A line plot is used to represent continuous data and is very useful for showing trends and changes over time. It connects individual data points with straight lines, making it easy to observe increases or decreases. Creating a line plot involves importing the required library, preparing the dataset, plofling the values, and enhancing the graph with labels and titles for befler understanding. 1. Import matplotlib.pyplot 2. Define x and y values 3. Use plot() function 4. Add labels and title 5. Display using show()

```
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [5,10,15,20]
plt.plot(x,y, marker='o')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Line Plot")
plt.show()
```

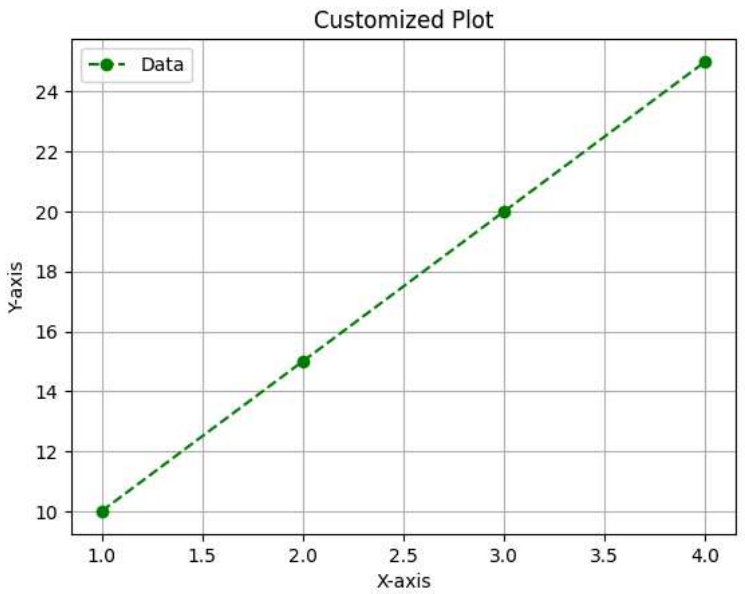



Q5. Why is customization important in data visualization? What customization options are available in Matplotlib?

ANS:-

Customization is important in data visualization because it enhances the clarity, readability, and presentation of the data. A well-customized graph helps users to focus on important information and makes the visualization more meaningful and aflractive. Without proper customization, graphs may look confusing or difficult to interpret. Therefore, customizing elements of a plot improves communication and understanding of the data. Matplotlib provides a wide range of customization options, allowing users to modify almost every part of a graph. Customization options include: 1. Changing colors, line styles, and markers 2. Adding titles and axis labels 3. Adjusting axis limits and scale 4. Modifying figure size5. Changing font style and size 6. Adjusting axis limits and scale

```
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [10,15,20,25]
plt.plot(x,y, color='green', linestyle='--', marker='o', label='Data')
plt.title("Customized Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.legend()
plt.grid()
plt.show()
```




Conclusion Data visualization is a fundamental part of data analysis that simplifies complex data and makes insights easier to understand. Tools like Matplotlib provide powerful capabilities to create a wide range of visualizations, from basic plots to highly customized graphs. By using visualization techniques during exploratory data analysis, analysts can detect patterns, trends, and anomalies efficiently. Overall, effective visualization combined with proper customization enhances both data interpretation and communication, making it an essential skill in data science and analytics.